Day 4 — Lab 4.1: CSV to Parquet using Azure Data Factory

Lab 4.1
Beginner–Intermediate

1. Business scenario

Your organization receives daily retail order data in CSV format and stores it in Azure Data Lake Storage Gen2.

The analytics team needs the data in Parquet format because it is more suitable for analytical processing and efficient storage.

You are assigned to create an Azure Data Factory pipeline that:

- Reads an incoming CSV file from ADLS Gen2.
- Converts the file into Parquet format.
- Stores the output in a separate processed folder.
- Validates the number of rows copied successfully.

2. Expected architecture

Azure Data Lake Storage Gen2

Raw zone

`raw/orders.csv`

Azure Data Factory — Copy Activity

Azure Data Lake Storage Gen2

Processed zone

`processed/orders.parquet`

3. Resources required

Each participant creates their own resources inside the existing shared training Resource Group.

Resource	Example name
Storage Account	`stadfuser01d4`
Data Factory	`adf-user01-d4`

Replace `user01` with the assigned participant suffix.

Storage configuration

- Performance: Standard
- Redundancy: Locally-redundant storage (LRS), if available and approved for the training region
- Account kind: StorageV2
- Hierarchical namespace: Enabled
- Filesystems: `raw`, `processed`

We will reuse these Day 4 resources in Labs 4.2 and 4.3 rather than creating them again.

4. Prerequisites

- Access to the shared Resource Group.
- Permission to create Storage Accounts and Data Factory resources.

- Permission to assign Azure RBAC roles to managed identities.
- Azure Portal access.

For storage access, the Data Factory managed identity will need Storage Blob Data Contributor on the storage account or the required filesystems. Creating this role assignment requires an appropriate role-assignment permission; ordinary Contributor access alone is insufficient.

5. Create the Storage Account

Azure Portal → Storage accounts → Create

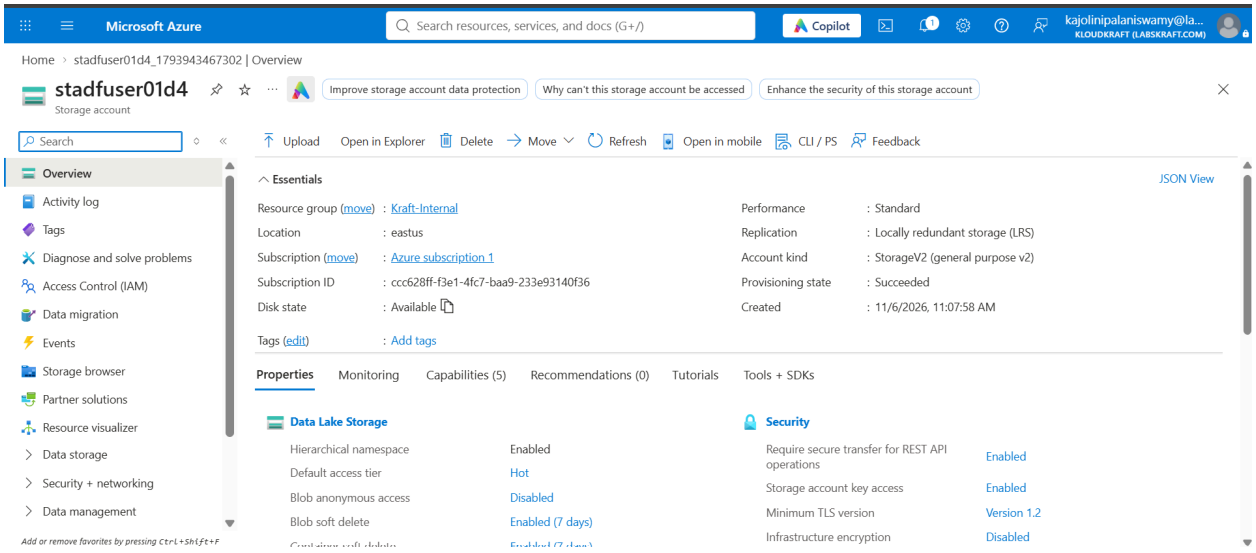
Configure the Basics tab:

Setting	Value
Subscription	Your training subscription
Resource Group	Existing shared RG
Storage account name	`stadfuser01d4`
Region	Approved training region
Performance	Standard
Redundancy	LRS

Under Advanced:

- Enable Hierarchical namespace.
- Leave other settings at their defaults unless the training environment requires otherwise.

Select Review + create → Create.

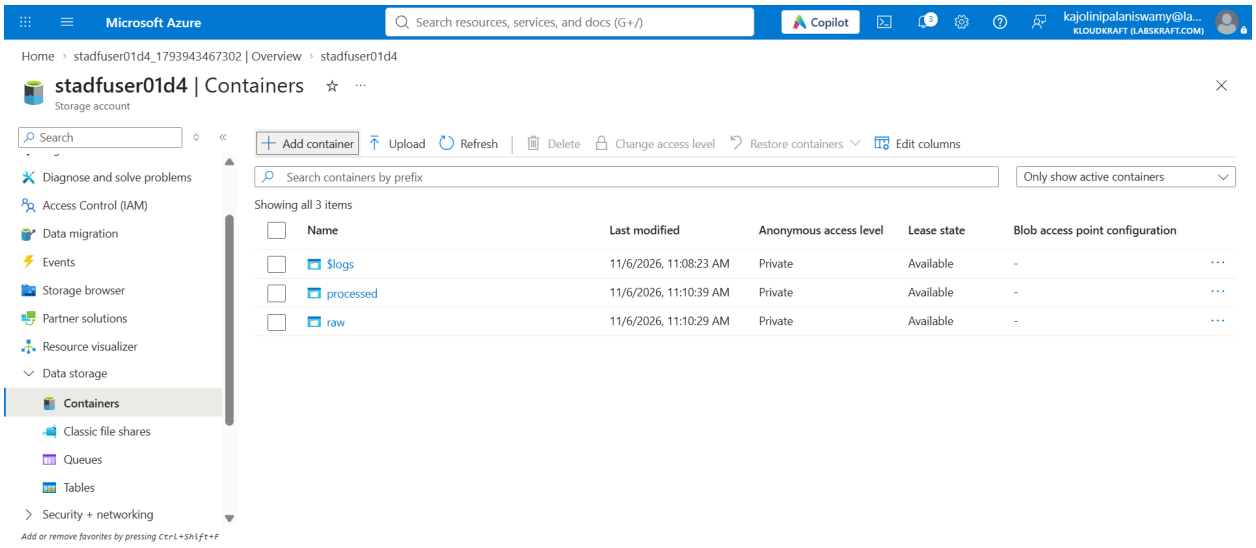


Important: The storage account name must be globally unique and lowercase.

Create the filesystems

1. Open the newly created Storage Account.
2. Select Data storage → Containers.
3. Select + Container.
4. Create the container `raw`.
5. Repeat and create `processed`.

These are ADLS Gen2 filesystems in the storage interface.



Create the folders

Open `raw` and create:

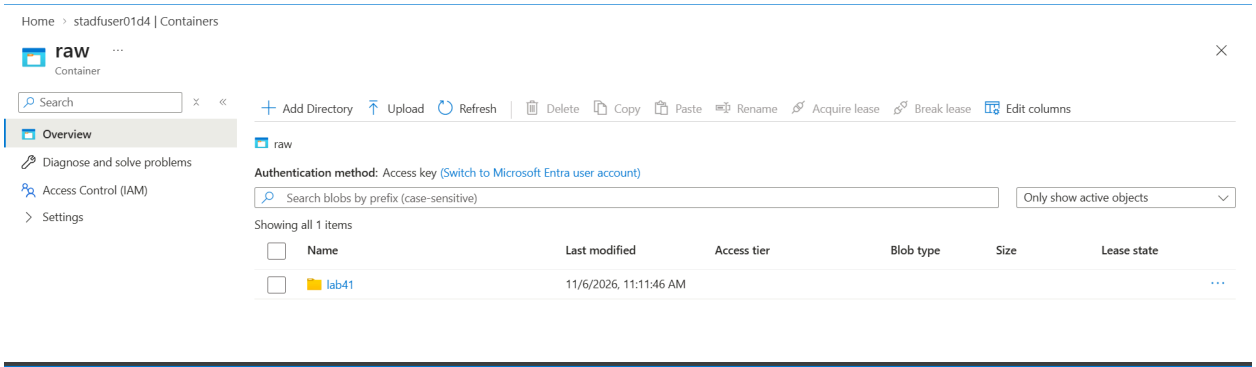
```
raw/
└── lab41/
```

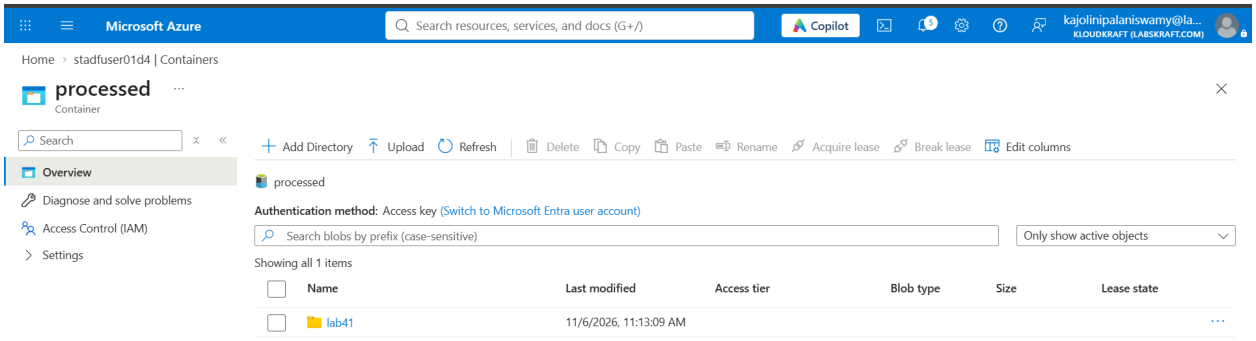
Open `processed` and create:

```
processed/
└── lab41/
```

Checkpoint 1

Both filesystems and their respective folders should now exist.





6. Create Azure Data Factory

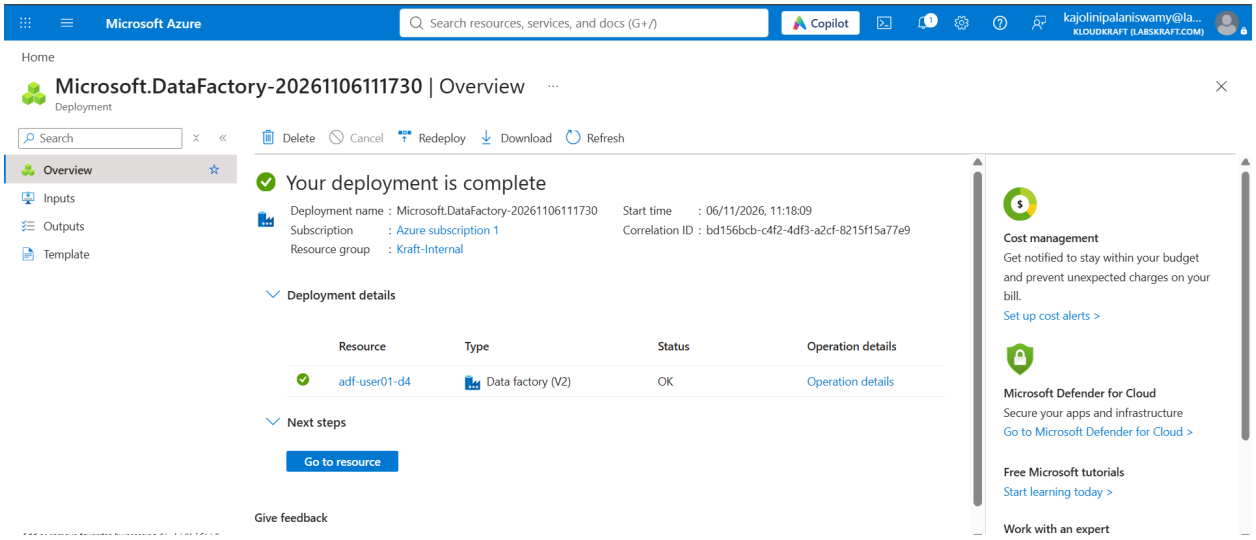
Azure Portal → Create a resource → Analytics → Data Factory

Configure:

Setting	Value
Name	`adf-user01-d4`
Subscription	Training subscription

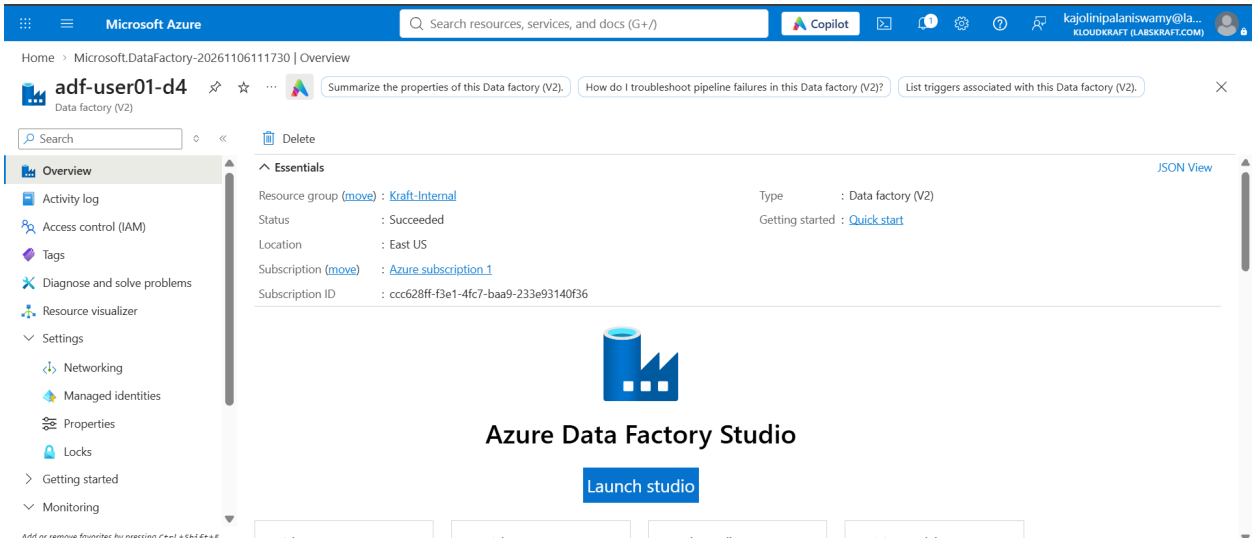
Resource Group	Existing shared RG
Region	Approved training region
Version	V2

Select Review + create → Create.



After deployment:

1. Open the Data Factory resource.
2. Select Launch Studio.
3. Wait for Azure Data Factory Studio to load.



Microsoft's portal tutorial follows this same resource creation and Studio launch workflow.

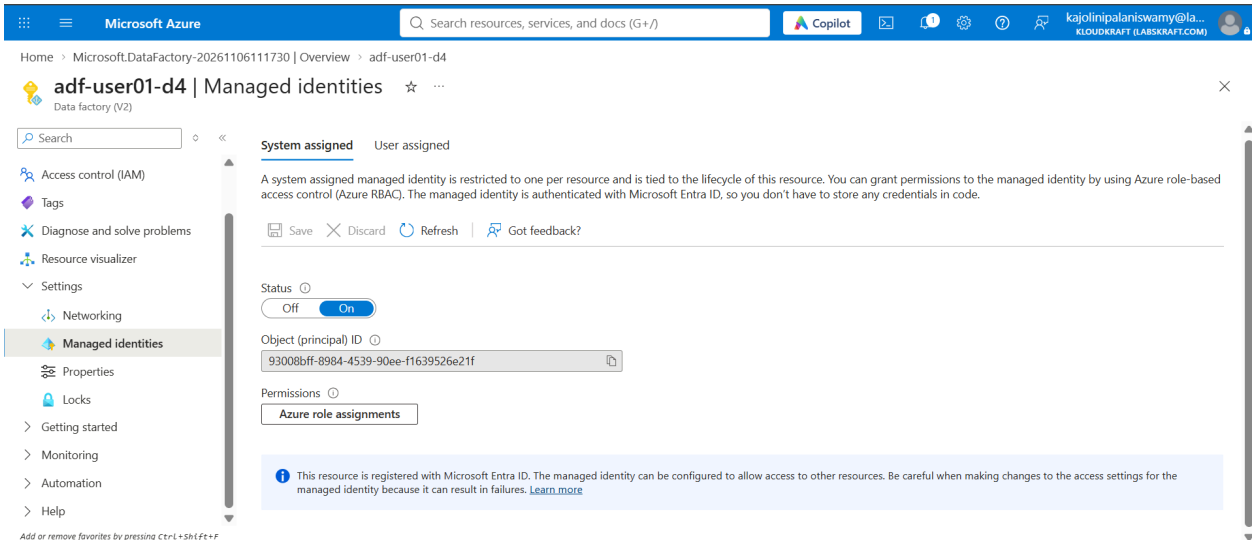
Microsoft Learn

7. Configure Data Factory Managed Identity

We will use Managed Identity instead of storing storage account keys in ADF.

Azure Portal → Data Factory → Identity

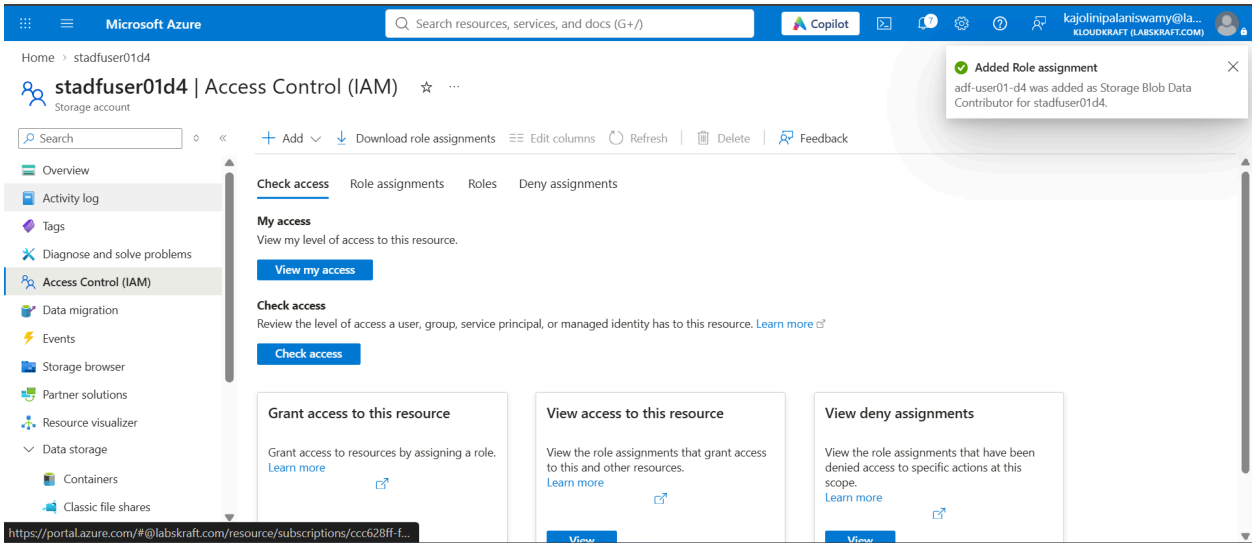
1. Open the System assigned tab.
2. Set Status to On, if it is not already enabled.
3. Select Save.
4. Confirm the identity has been created.



Now grant storage access:

1. Open your Storage Account.
2. Select Access Control (IAM).
3. Select Add → Add role assignment.
4. Choose `Storage Blob Data Contributor`.
5. Select the Managed identity option.
6. Choose your Data Factory.
7. Select Review + assign.

Wait for the role assignment to propagate.



Checkpoint 2
Data Factory's managed identity has access to the ADLS Gen2 storage account.

8. Create the ADLS Gen2 Linked Service

Open Azure Data Factory Studio.

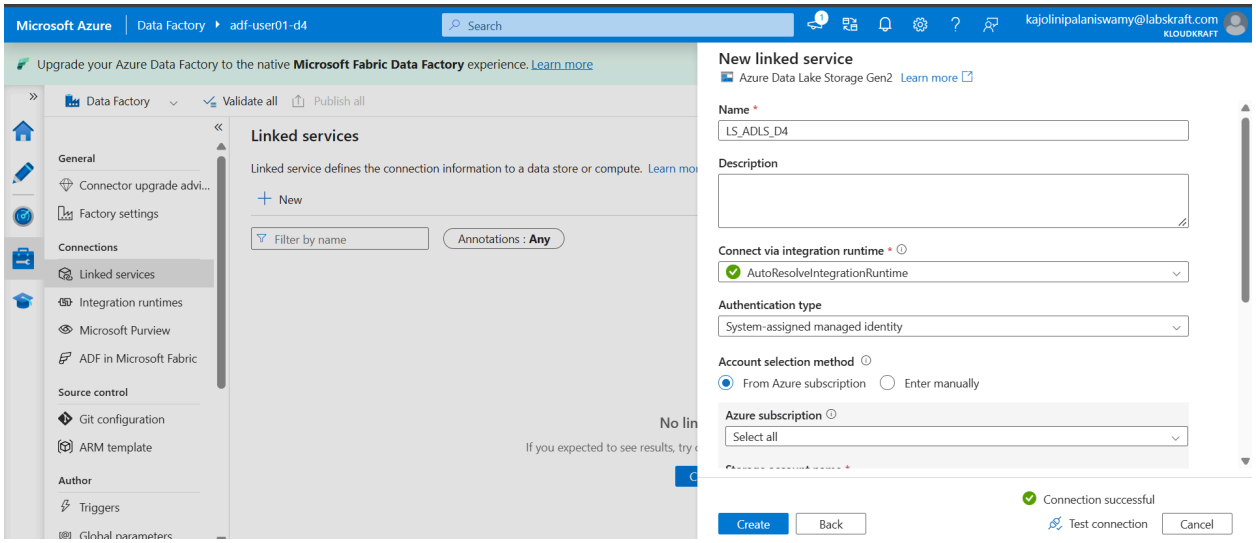
1. Select Manage from the left navigation.
2. Open Linked services.
3. Select + New.
4. Search for `Azure Data Lake Storage Gen2`.
5. Select it and click Continue.

Configure:

Setting	Value
Name	`LS_ADLS_D4`
Integration Runtime	AutoResolveIntegration Runtime
Authentication	Managed Identity
Account selection	Your Day 4 Storage Account

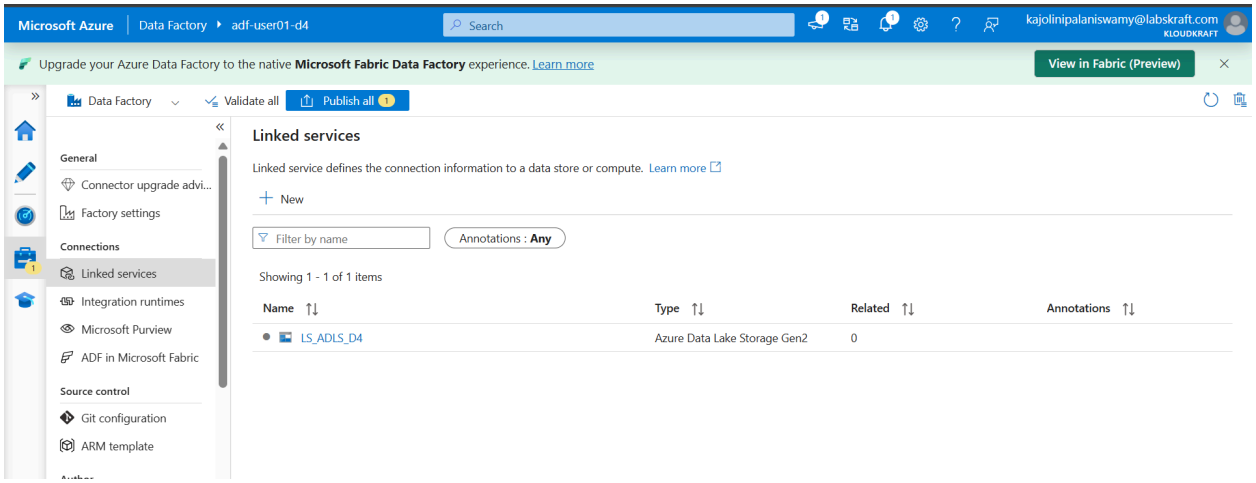
Select Test connection.

Expected result
Connection successful.



Select Create.

ADF supports managed identity authentication for ADLS Gen2 linked services.



Microsoft Learn+1

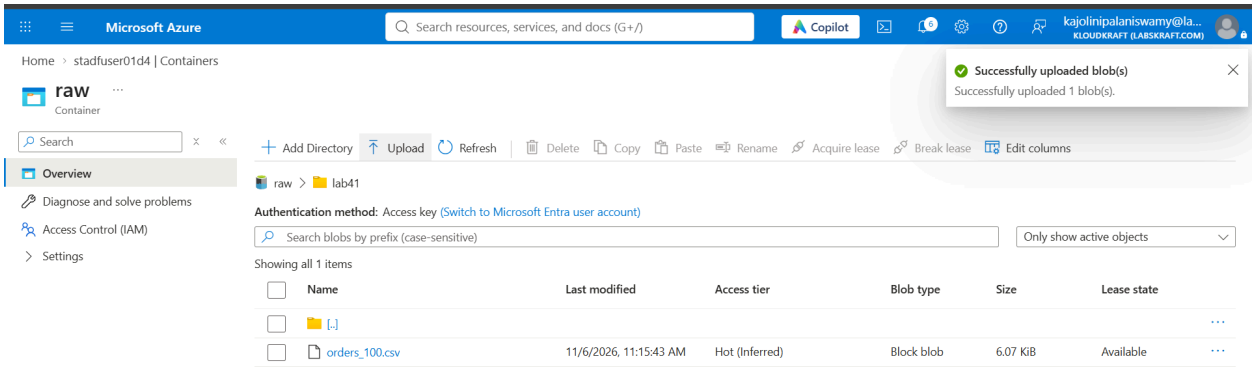
9. Upload the input CSV file

For this lab, our planned dataset is `orders_100.csv` containing 100 order records.

1. Open Storage Account → Containers → `raw`.
2. Open the `lab41` folder.
3. Select Upload.
4. Select `orders_100.csv`.
5. Select Upload.

Expected storage path:

```
raw/lab41/orders_100.csv
```



We'll generate the actual CSV artifact after finalizing all three Day 4 labs.

10. Create the source dataset

Open ADF Studio → Author (pencil icon).

1. Select the + button.
2. Select Dataset.
3. Search for `Azure Data Lake Storage Gen2`.
4. Select it and click Continue.
5. Choose DelimitedText and click Continue.

Configure the dataset:

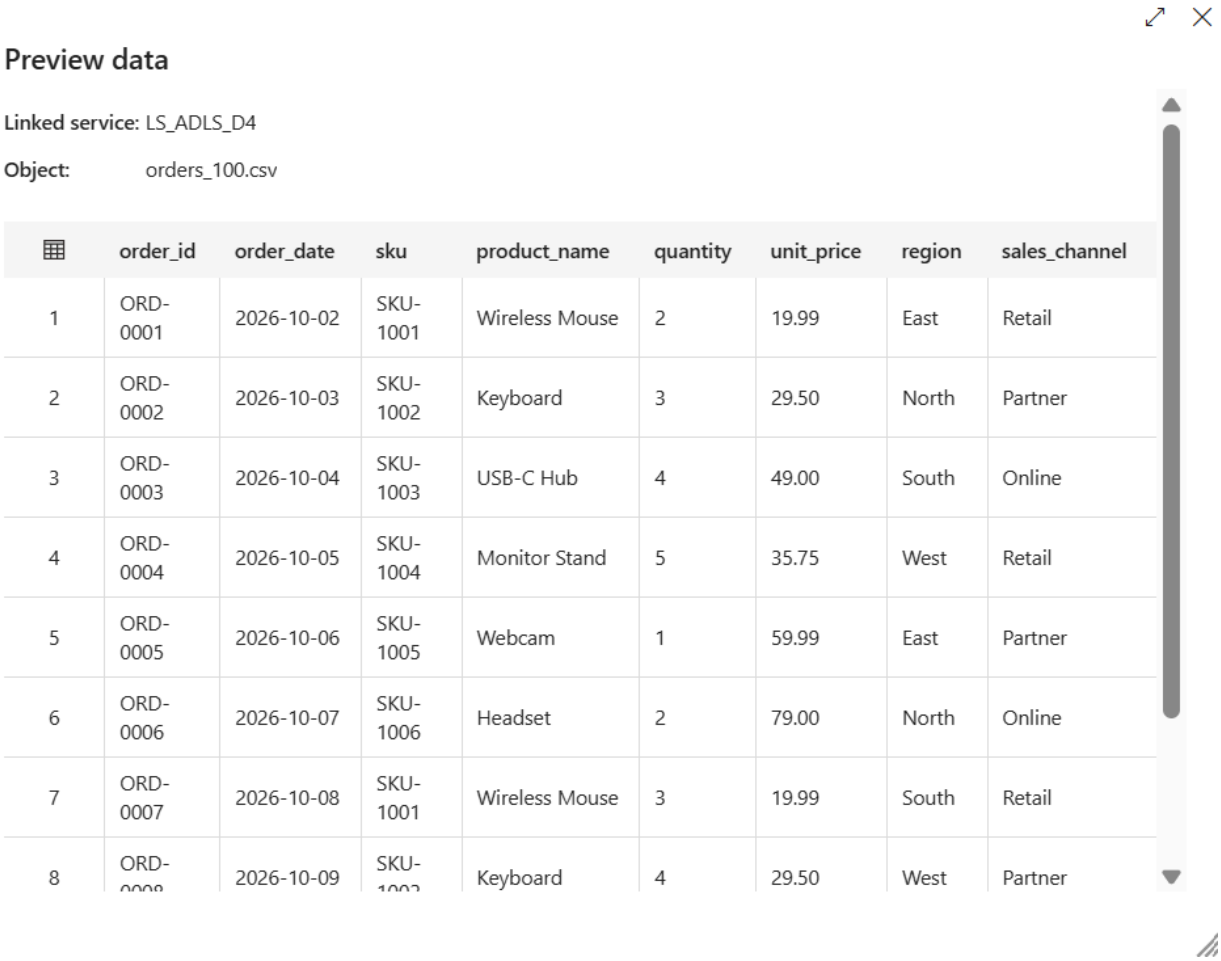
Setting	Value
Name	`DS_D4_Orders_CSV`
Linked service	`LS_ADLS_D4`
Filesystem	`raw`
Directory	`lab41`
File	`orders_100.csv`
First row as header	Enabled
Column delimiter	Comma
Encoding	UTF-8

Select OK.

Open the dataset and select Preview data.

Expected result:

- CSV records are displayed.
- Column names appear correctly.
- No connection or permission errors.



Preview data

Linked service: LS_ADLS_D4

Object: orders_100.csv

	order_id	order_date	sku	product_name	quantity	unit_price	region	sales_channel
1	ORD-0001	2026-10-02	SKU-1001	Wireless Mouse	2	19.99	East	Retail
2	ORD-0002	2026-10-03	SKU-1002	Keyboard	3	29.50	North	Partner
3	ORD-0003	2026-10-04	SKU-1003	USB-C Hub	4	49.00	South	Online
4	ORD-0004	2026-10-05	SKU-1004	Monitor Stand	5	35.75	West	Retail
5	ORD-0005	2026-10-06	SKU-1005	Webcam	1	59.99	East	Partner
6	ORD-0006	2026-10-07	SKU-1006	Headset	2	79.00	North	Online
7	ORD-0007	2026-10-08	SKU-1001	Wireless Mouse	3	19.99	South	Retail
8	ORD-0008	2026-10-09	SKU-1002	Keyboard	4	29.50	West	Partner

11. Create the destination dataset

Repeat the dataset creation process.

1. Select + → Dataset.
2. Choose Azure Data Lake Storage Gen2.
3. Select Parquet.

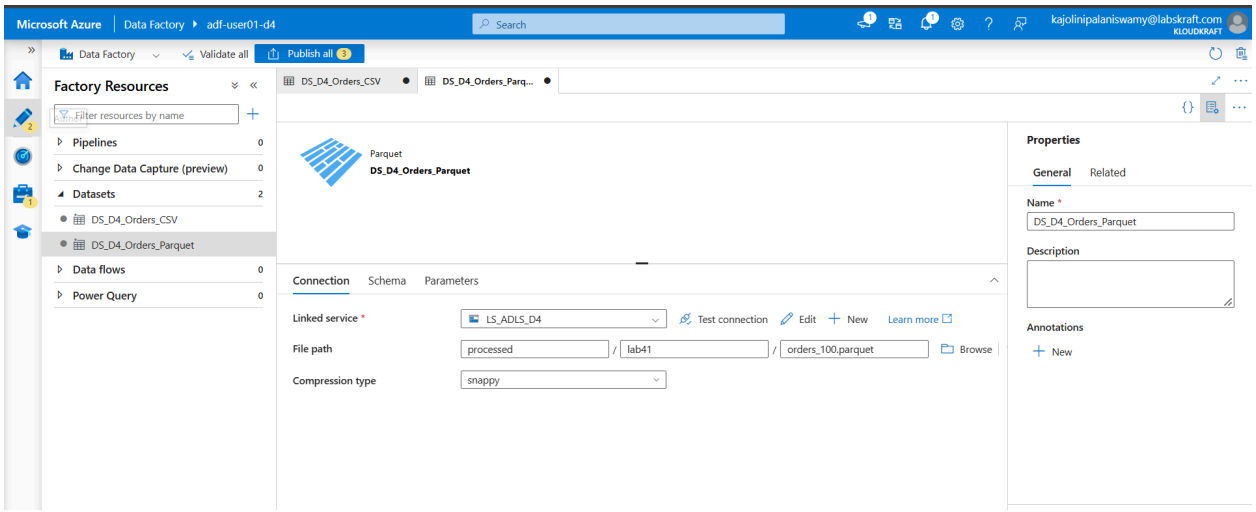
4. Click Continue.

Configure:

Setting	Value
Name	`DS_D4_Orders_Parquet`
Linked service	`LS_ADLS_D4`
Filesystem	`processed`
Directory	`lab41`
File	`orders_100.parquet`

Select OK.

The destination dataset is now ready.



12. Create the ADF pipeline

Navigate to Author → + → Pipeline.

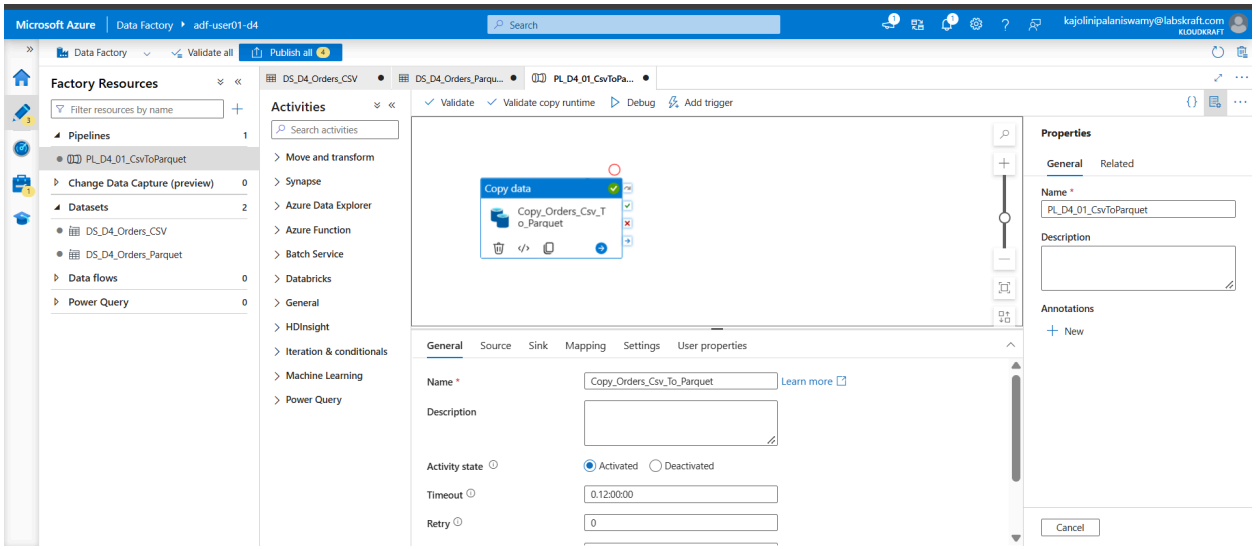
Set the pipeline name:

```
PL_D4_01_CsvToParquet
```

From the Activities pane:

1. Expand Move & Transform.
2. Drag Copy data onto the pipeline canvas.
3. Select the activity.
4. Under General, set its name to:

```
Copy_Orders_Csv_To_Parquet
```



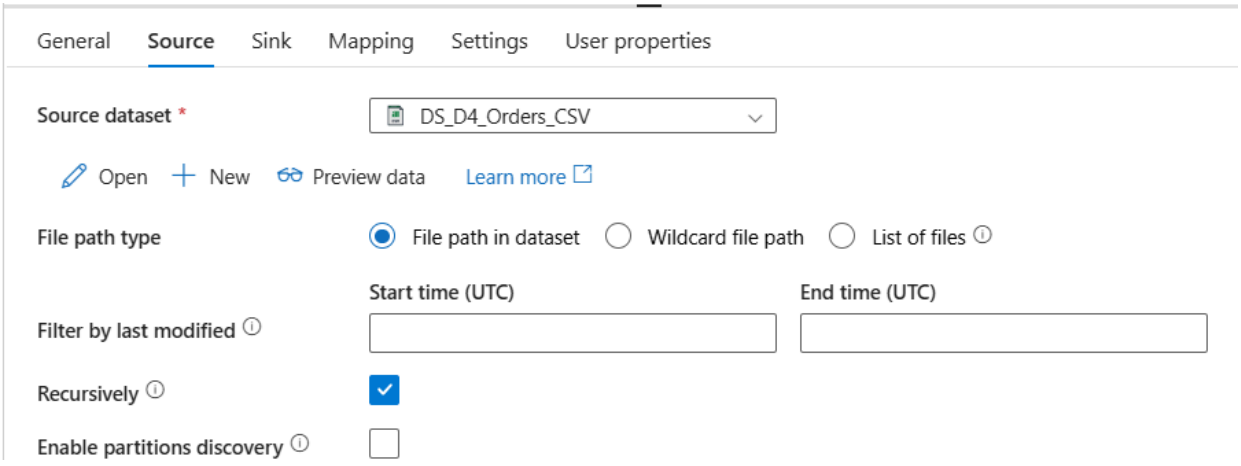
13. Configure the Copy Activity

Select the Copy Activity.

Source tab

Setting	Value
Source dataset	DS_D4_Orders_CSV

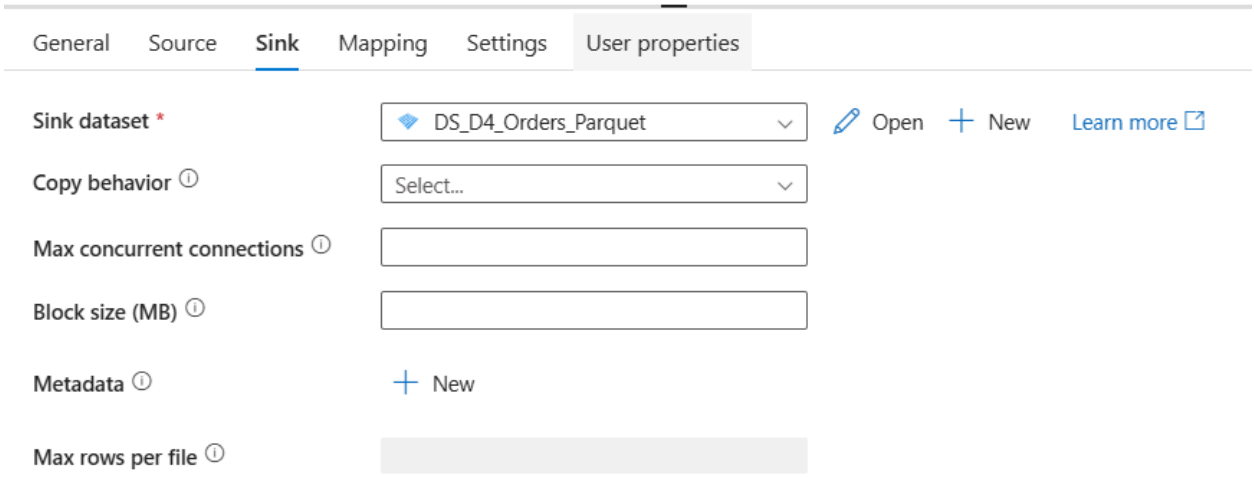
Confirm the correct source file is selected.



Sink tab

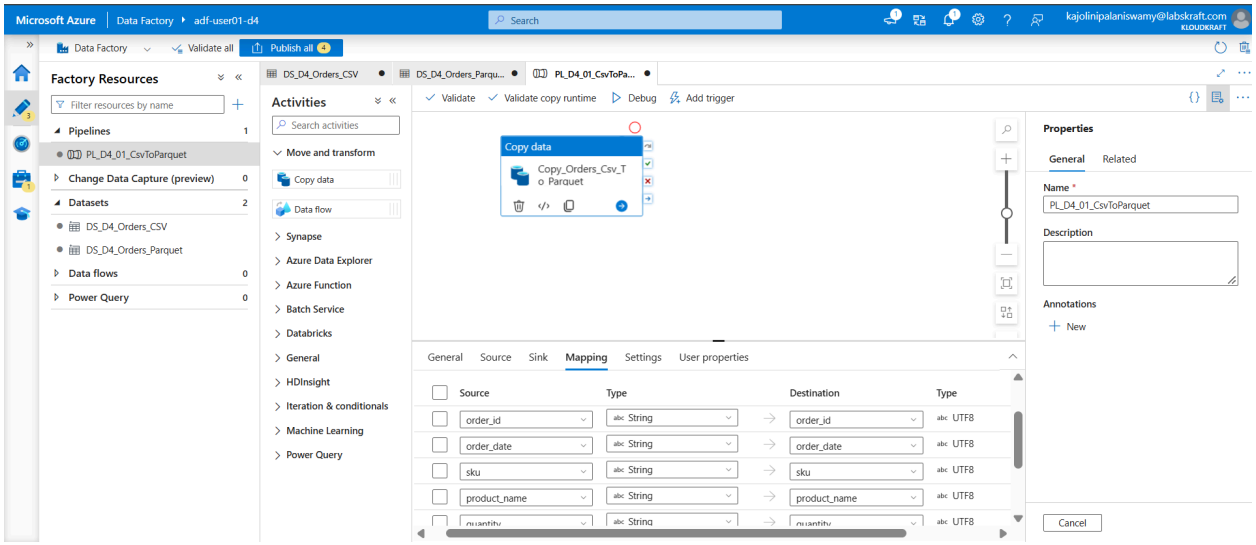
Setting	Value
Sink dataset	`DS_D4_Orders_Parquet`

Keep the remaining settings at their defaults.



Mapping tab

1. Select Import schemas if available.
2. Confirm all eight source columns are mapped to the destination.
3. Verify there are no unexpected missing columns.

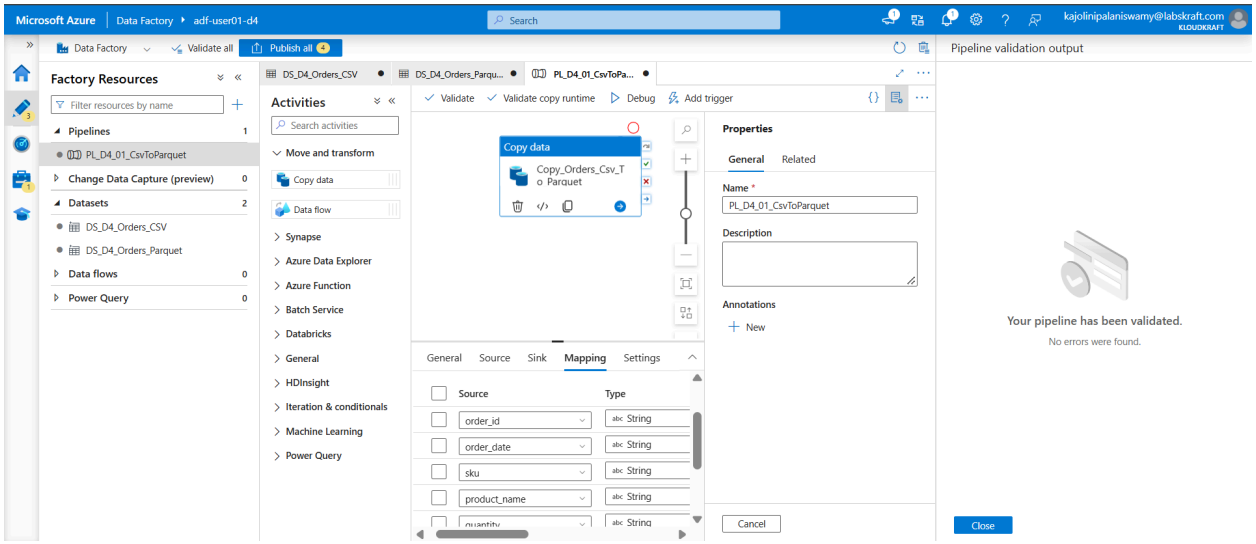


The pipeline should now look like this:

[Use private endpoints to create an Azure Data Factory pipeline - Azure Data Factory | Microsoft Learn
learn.microsoft.com]

14. Validate and execute the pipeline

1. Select Validate from the pipeline toolbar.
2. Resolve any validation errors.

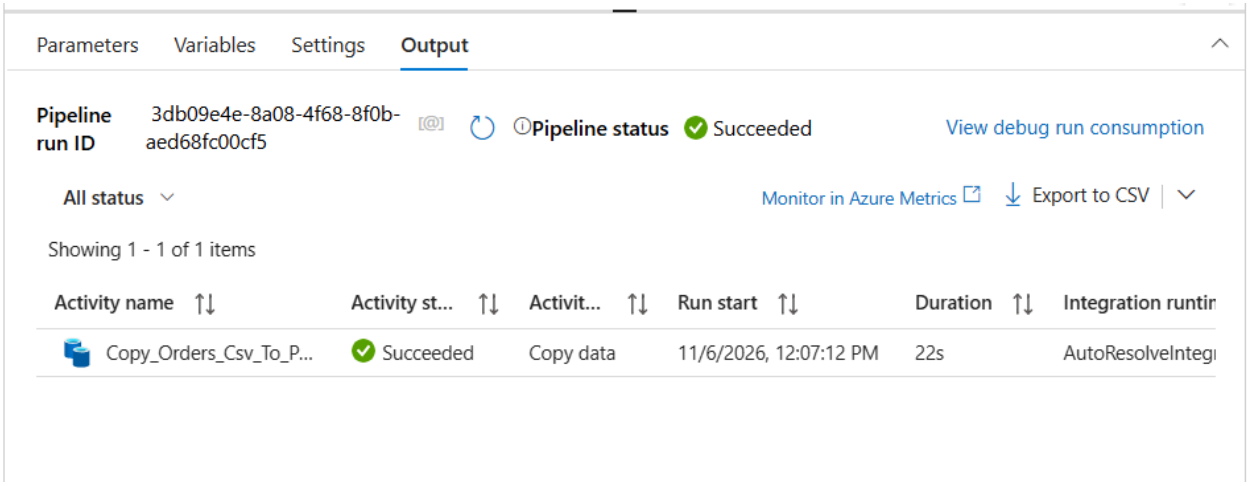


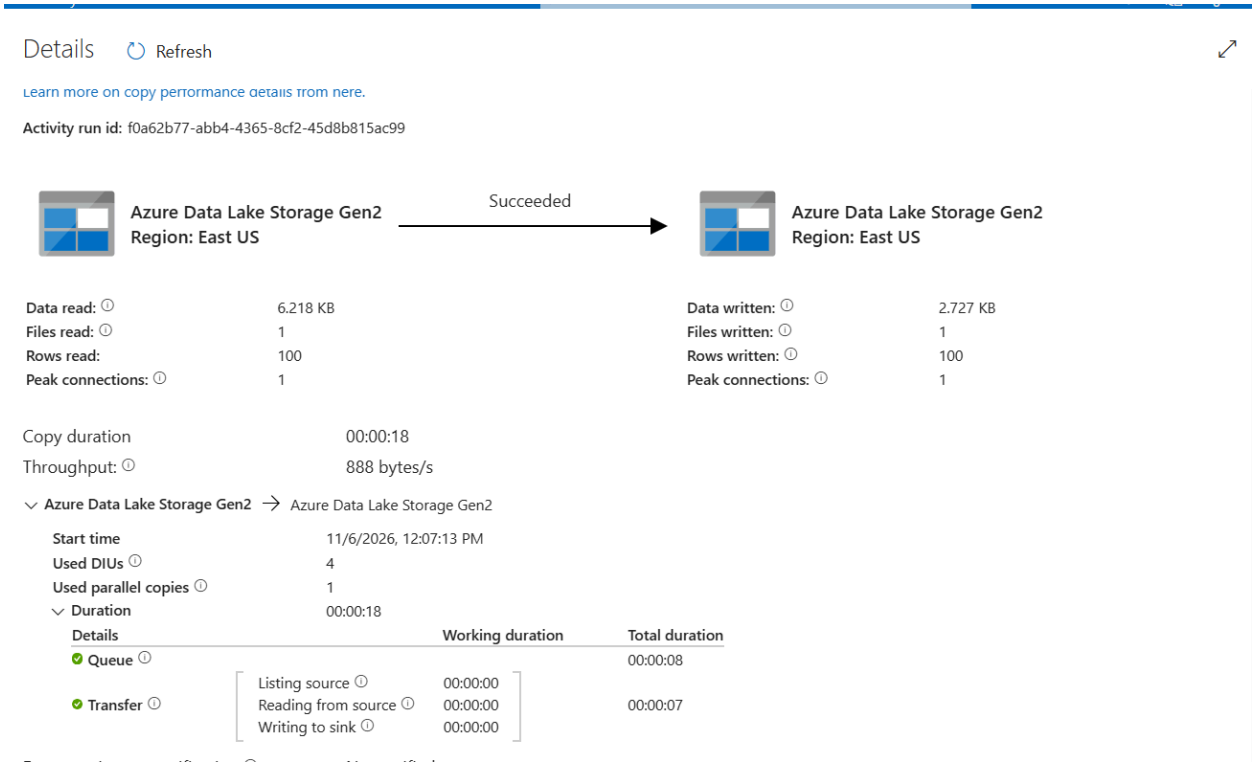
3. Select Debug.
4. Wait for the Copy Activity execution to finish.
5. Open the Output panel.

Expected result

```
None
Pipeline status: Succeeded
Copy activity: Succeeded
Rows read: 100
Rows copied: 100
```

The exact metrics will be confirmed against the generated dataset and actual execution.





15. Verify the output in ADLS Gen2

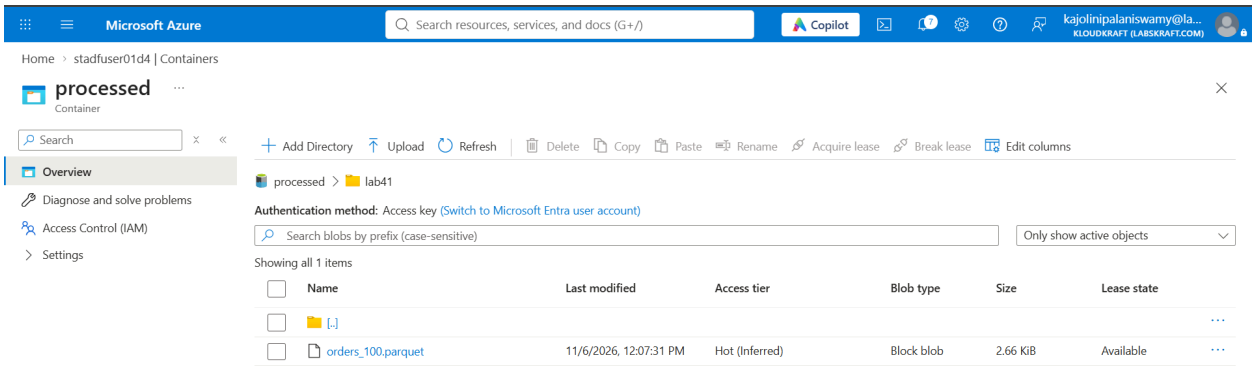
Return to Azure Portal.

1. Open the Day 4 Storage Account.
2. Select Containers → `processed`.
3. Open `lab41`.

Expected output:

```
None
processed/
└── lab41/
    └── orders_100.parquet
```

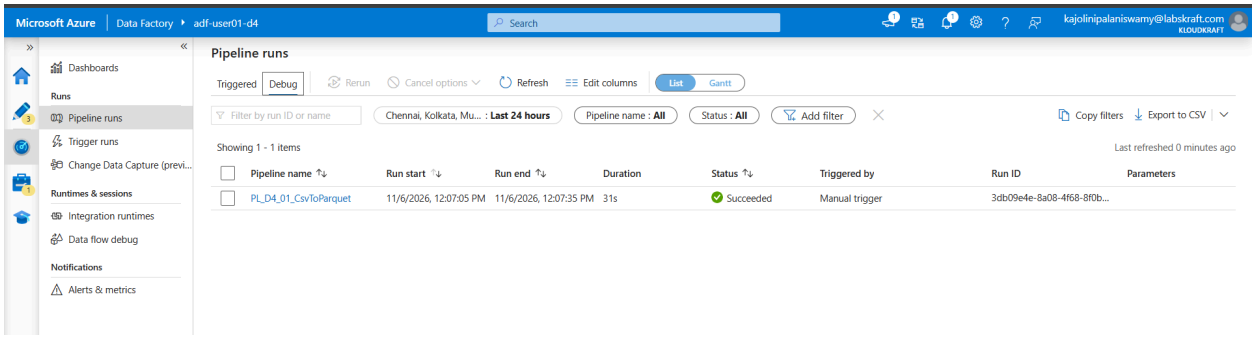
Confirm that the Parquet file exists.



16. Monitor the pipeline execution

Open ADF Studio → Monitor.

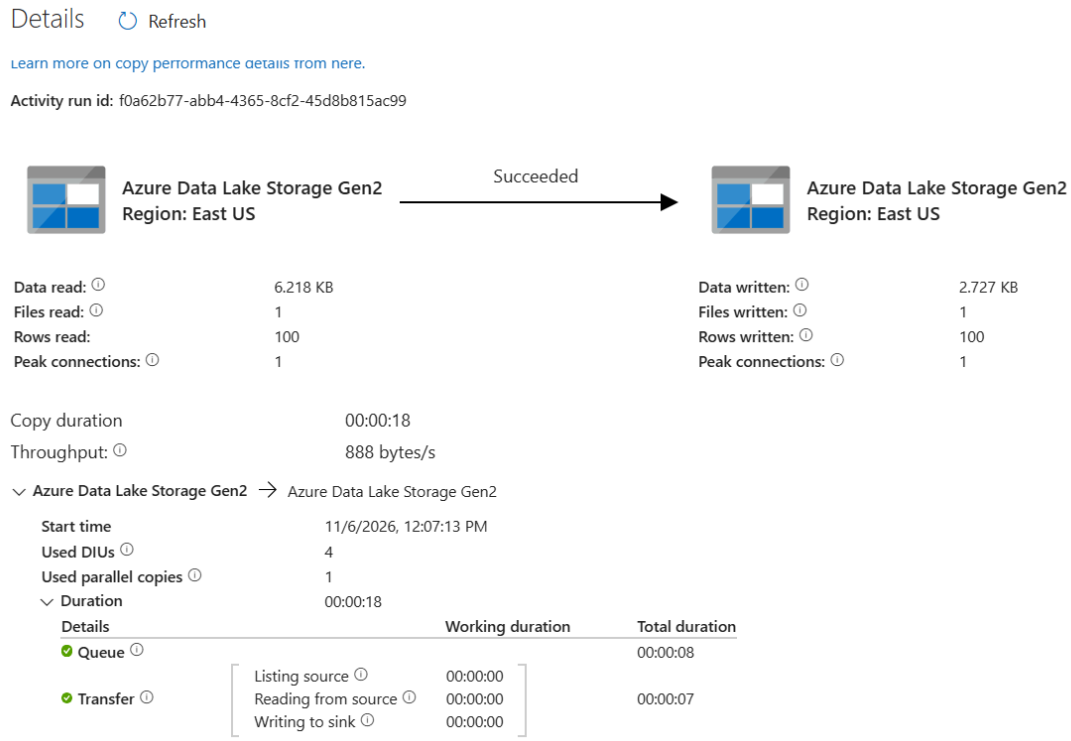
1. Open Pipeline runs.
2. Locate `PL_D4_01_CsvToParquet`.
3. Confirm the status is Succeeded.



4. Open the pipeline run.
5. Inspect the Copy Activity details.
6. Review rows read, rows copied, duration and output details.

ADF provides activity-level monitoring and Copy Activity execution metrics.

Microsoft Learn+1



17. Evidence collection

Capture these screenshots:

- Storage Account with Hierarchical namespace enabled.
- `raw/lab41` containing the input CSV.
- `processed/lab41` containing the Parquet file.
- ADF linked service successful connection test.
- Source and destination dataset configurations.
- Pipeline canvas showing Copy Activity.
- Successful pipeline execution.
- Copy Activity output metrics.

18. Lab 4.1 completion criteria

Validation	Expected result
ADLS Gen2 setup	PASS
ADF linked service	Connected
Source dataset	CSV preview successful
Destination dataset	Parquet configured
Pipeline validation	Successful
Pipeline execution	Succeeded
Input records	100
Output records	100
Parquet file	Available

Cleanup: Do not delete the Storage Account or Data Factory after Lab 4.1. We will reuse both in Labs 4.2 and 4.3. Clean up only at the end of Day 4.